DEBRIEF · SIMULATED CASE

Stab wound: tamponade, PEA and pericardiocentesis

M 30 y 80 kg · 11:31 simulated (3 min 46 s on the clock) · 10/19/2026, 5:00:58 AM · Run by: Showcase 2, 24 October

Key times

Recognition (instructor's tick)
1 min 07 s
00:00 → 01:07

Start → first team action
7 min 51 s
00:00 → 07:51 (CPR, ventilated with FiO_2 1.0)

Start CPR within 10 s of the pulse loss
3 s
07:48 → 07:51 · asked ≤ 10 s

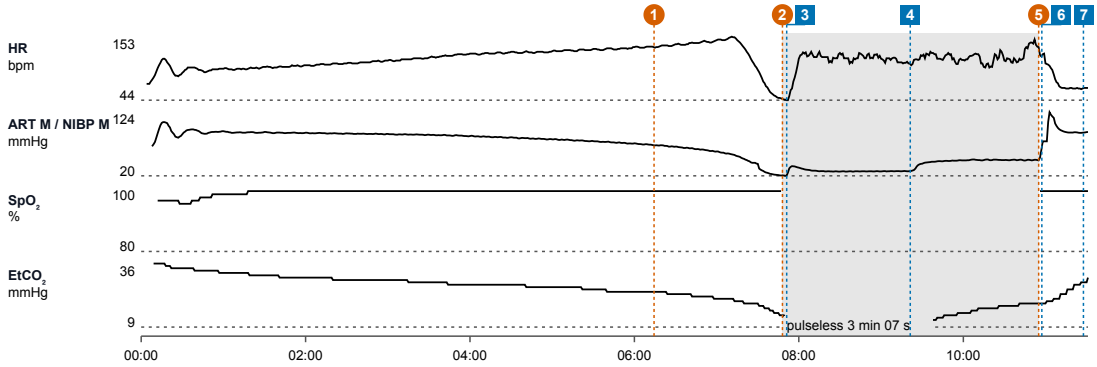
Pericardiocentesis within 120 s of starting CPR
1 min 30 s
07:51 → 09:21 · asked ≤ 2 min 00 s

Stop CPR (“Pulse back”) within 30 s of the pulse…
2 s
10:55 → 10:57 · asked ≤ 30 s

Pulseless in all
3 min 07 s
first lost 07:48

The course

■ team action · ● patient event · | scenario state · numbers match Key moments



Objectives

What the record shows: it reads the run, it does not grade the person

1. ✓ Observed — Name the tamponade within 180 s of arrival from the monitor: tachycardia, narrow pulse pressure, CVP above 15 mmHg (instructor-assessed)
   Observed by the instructor at 01:07.
2. ✓ Observed — Start CPR within 10 s of the pulse loss — 3 s of 10 s
   CPR 3 s after the pulse loss (07:48 → 07:51; within 10 s asked).
3. ✓ Observed — Pericardiocentesis within 120 s of starting CPR — 1 min 30 s of 2 min 00 s
   The state "CPR: pericardium drained, catheter left in" 1 min 30 s after CPR (07:51 → 09:21; within 2 min 00 s asked).
4. ✓ Observed — Stop CPR (“Pulse back”) within 30 s of the pulse returning on the arterial line and the capnogram — 2 s of 30 s
   The state "Return of circulation: ventilated, catheter in the pericardium" 2 s after the return of circulation (10:55 → 10:57; within 30 s asked).
5. ◆ Discuss — Better: pericardiocentesis before the pulse loss, while the systolic pressure is above 60 mmHg — then no PEA follows
   The pulse loss at 07:48

Key moments and what the monitor showed

values at that second

n	Time	Who	What happened	HR	MAP	SpO_2	$EtCO_2$
1	06:14	Patient	Deteriorating: systolic below 80	135	70	99	24
	Alarms since the last row: CVP high, SpO_2: low perfusion, ART S low						
2	07:48	Patient	Pulse lost (the model) · PEA	46	21	–	14
	Alarms since the last row: ART M low, HR high, ART D low, $EtCO_2$ low, ART: no pulsatile pressure, HR low						
3	07:51	Team	CPR, ventilated with FiO_2 1.0	44	20	–	13
4	09:21	Team	CPR: pericardium drained, catheter left in	107	28	–	9
	Alarms since the last row: ART S low, ART D low						
5	10:55	Patient	Circulation returned (the model)	135	45	–	19
6	10:57	Team	Return of circulation: ventilated, catheter in the pericardium	119	63	99	19
7	11:27	Team	Post-ROSC: 500 mL crystalloid running	64	90	99	28
	Alarms since the last row: ART D high, ART M high						

SpO_2 –: no reliable oximeter reading without a pulse (during CPR the pleth is compression artefact).

Measured facts

Lowest SpO_2	95 % at 00:28
Lowest ART M / NIBP M	20 mmHg at 07:51
Highest $EtCO_2$	36 mmHg at 00:09
Highest T1 / T2	36.8 °C at 00:01
Pulseless	3 min 07 s in all; first lost at 07:48
Return of circulation	3 min 07 s after the pulse loss (10:55)
Labs (the model): last (lowest–highest)	K^+ 4.2 mmol/L (4.2–4.2) · Lactate 2.2 mmol/L (1.0–2.2) · pH 7.28 (7.28–7.41) · Glucose 5.8 mmol/L (5.6–5.8) · Tcore 36.8 °C (36.8–36.8)

Talking points

Reactions
How did that feel? What stands out for you?

Description
In a sentence or two: what was the problem, and when did you know?

Analysis
Take one key moment: what did you see, what did you think, what did you do?

Summary
One thing to keep doing, one thing to do differently next time.

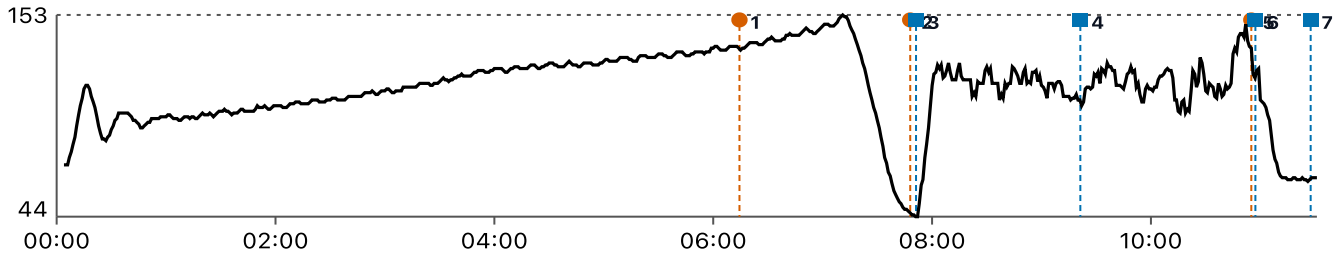
Notes

Kit round Q2 rehearsal on the rc.2 release build.

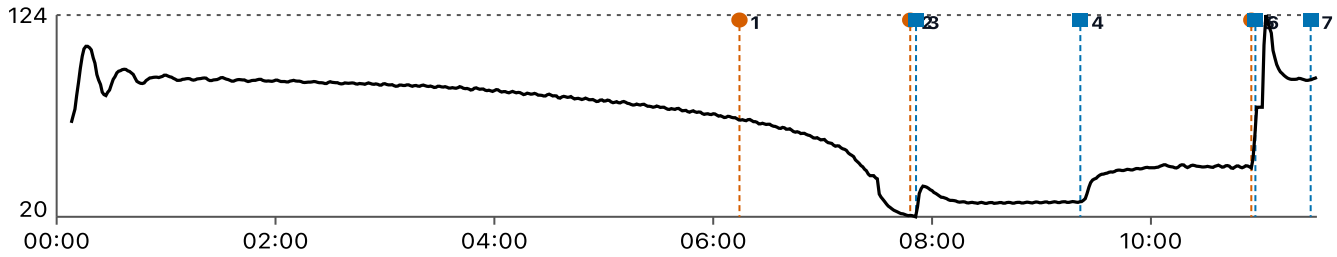
Trends

team action · patient event · | scenario state

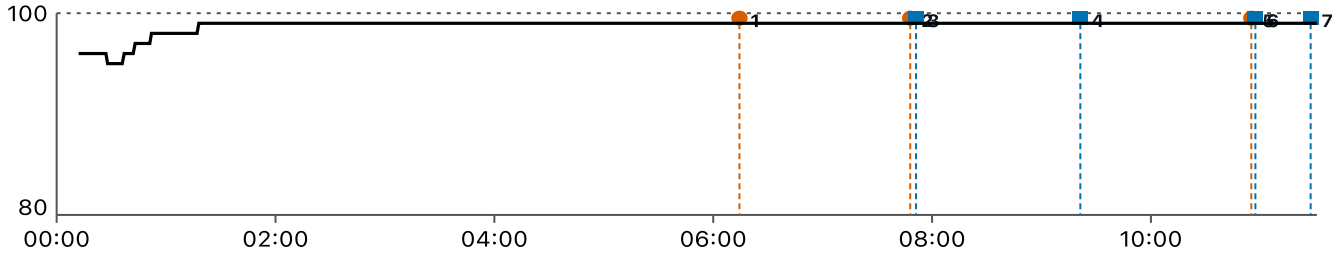
HR bpm



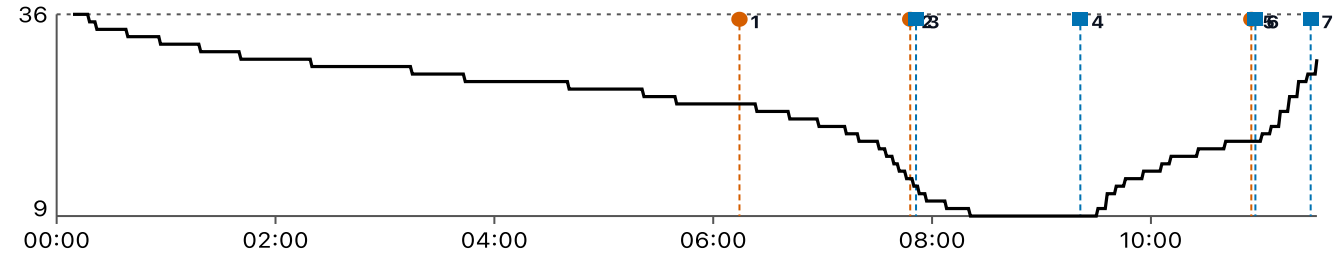
ART M / NIBP M mmHg



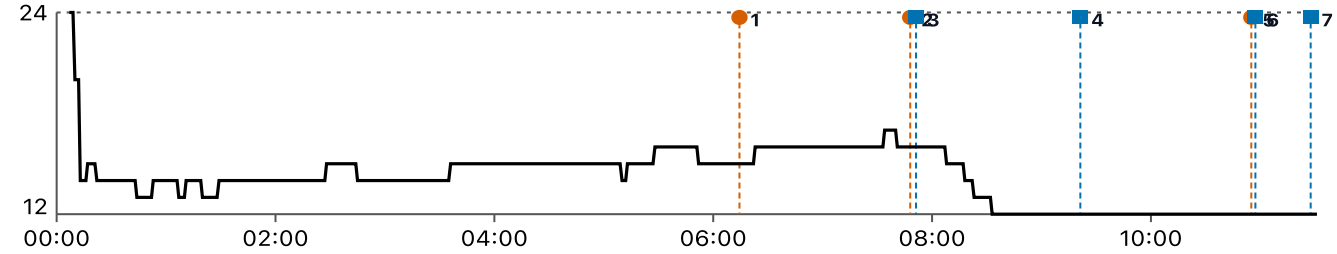
SpO_2 %

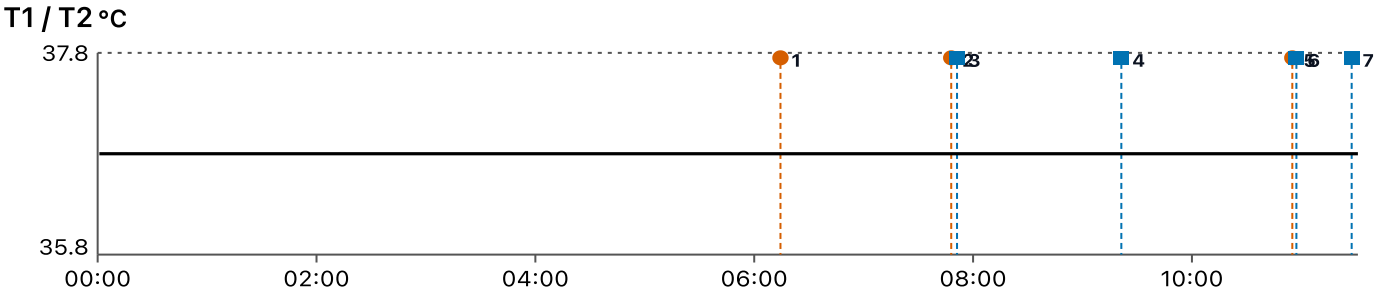


$EtCO_2$ mmHg



awRR / RR /min





Drugs and fluids

Time	Given
11:27	Crystalloid 500 mL over 5 min

Timeline

Time	Kind	Event
00:00	Scenario	Scenario loaded: Stab wound: tamponade, PEA and pericardiocentesis
00:00	Scenario	Scenario state: Arrival: tachycardic, narrow pulse pressure, raised CVP
00:00	Scenario	Breathing spontaneously
00:00	Scenario	Cardiac tamponade: severe
00:02	Action	All alarms on
00:03	Action	Speed ×8
00:06	Bookmark	Bookmark: Bookmark 1 at 00:06
00:22	Alarm	CVP high
01:07	Note	Objective met: Name the tamponade within 180 s of arrival from the monitor: tachycardia, narrow pulse pressure, CVP above 15 mmHg (instructor-assessed)
05:59	Alarm	SpO_2: low perfusion
06:06	Alarm	ART S low
06:14	Scenario	Scenario state: Deteriorating: systolic below 80
07:03	Alarm	ART M low
07:03	Action	Speed ×1
07:10	Alarm	HR high
07:17	Alarm	ART D low
07:21	Alarm	$EtCO_2$ low
07:31	Alarm	ART: no pulsatile pressure
07:45	Alarm	HR low
07:48	Patient	Pulse lost (the model)
07:49	Scenario	Scenario state: PEA: organised rhythm, no pulse; the pericardium holds about 370 mL
07:49	Scenario	Cardiac tamponade: severe
07:51	Scenario	Scenario state: CPR, ventilated with FiO_2 1.0
07:51	Scenario	CPR started, 110 /min
07:51	Scenario	Tracheal tube in place

Time	Kind	Event
07:51	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 100 %
07:52	Action	Speed ×4
07:51	Patient	Ventilated breaths began
07:57	Alarm	ART S low
07:57	Alarm	ART D low
09:02	Action	Speed ×1
09:21	Scenario	Scenario state: CPR: pericardium drained, catheter left in
09:21	Scenario	Cardiac tamponade: severe
09:22	Action	Speed ×4
10:21	Action	Speed ×1
10:55	Patient	Circulation returned (the model)
10:57	Scenario	Scenario state: Return of circulation: ventilated, catheter in the pericardium
10:57	Scenario	CPR stopped
11:04	Alarm	ART D high
11:04	Alarm	ART M high
11:27	Scenario	Scenario state: Post-ROSC: 500 mL crystalloid running
11:27	Scenario	Crystalloid 500 mL over 5 min

Bookmarks

- 00:06 Bookmark: Bookmark 1 at 00:06

Appendix: one row per minute

Time	HR bpm	ART M / NIBP M mmHg	SpO_2 %	$EtCO_2$ mmHg	awRR / RR /min	T1 / T2 °C	K^+ mmol/L	Lactate mmol/L	pH	Glucose mmol/L	Tcore °C
00:00	72	68	96	36	24	36.8	4.2	1.0	7.41	5.6	36.8
01:00	98	93	98	32	14	36.8	4.2	1.0	7.39	5.6	36.8
02:00	104	90	99	30	14	36.8	4.2	1.0	7.39	5.6	36.8
03:00	112	89	99	29	14	36.8	4.2	1.0	7.38	5.6	36.8
04:00	124	85	99	27	15	36.8	4.2	1.1	7.38	5.6	36.8
05:00	128	80	99	26	15	36.8	4.2	1.1	7.37	5.6	36.8
06:00	135	73	99	24	15	36.8	4.2	1.2	7.37	5.6	36.8
07:00	147	60	99	21	16	36.8	4.2	1.4	7.37	5.6	36.8
08:00	115	34	99	11	16	36.8	4.2	1.6	7.35	5.6	36.8
09:00	115	27	99	9	12	36.8	4.2	1.9	7.31	5.6	36.8
10:00	113	45	99	15	12	36.8	4.2	2.1	7.29	5.7	36.8
11:00	107	76	99	19	12	36.8	4.2	2.2	7.28	5.8	36.8